作業系統HW2

AI三 B1228005 胡樂麒

1.安裝所需套件，建立資料夾

```
logan@logan-VirtualBox:~$ sudo apt install -y build-essential libncurses-dev flex bison libssl-dev libelf-dev \
  dwarves bc fakeroot ccache wget curl git ca-certificates \
  python3
```

```
logan@logan-VirtualBox:~$ mkdir -p ~/kernel_build
cd ~/kernel_build
```

2.原始kernel版本為6.8.0-87-generic,使用apt source linux-hwe-6.8 下載並解壓縮

```
logan@logan-VirtualBox:~/kernel_build/linux-hwe-6.8-6.8.0$ uname -r
6.8.0-87-generic
logan@logan-VirtualBox:~/kernel_build/linux-hwe-6.8-6.8.0$
```

```
logan@logan-VirtualBox:~/kernel_build$ cd ~/kernel_build/linux-hwe-6.8-6.8.0
logan@logan-VirtualBox:~/kernel_build/linux-hwe-6.8-6.8.0$ cp -v /boot/config-$
(uname -r) .config
'/boot/config-6.8.0-87-generic' -> '.config'
logan@logan-VirtualBox:~/kernel_build/linux-hwe-6.8-6.8.0$ make oldconfig
  HOSTCC  scripts/basic/fixdep
  HOSTCC  scripts/kconfig/conf.o
  HOSTCC  scripts/kconfig/confdata.o
  HOSTCC  scripts/kconfig/expr.o
  LEX     scripts/kconfig/lexer.lex.c
  YACC    scripts/kconfig/parser.tab.[ch]
  HOSTCC  scripts/kconfig/lexer.lex.o
  HOSTCC  scripts/kconfig/menu.o
  HOSTCC  scripts/kconfig/parser.tab.o
  HOSTCC  scripts/kconfig/preprocess.o
  HOSTCC  scripts/kconfig/symbol.o
  HOSTCC  scripts/kconfig/util.o
  HOSTLD  scripts/kconfig/conf
#
# configuration written to .config
#
```

3. 編譯 Kernel, 安裝模組, 安裝kernel本體

make -j"$(nproc)" , sudo make modules_install, sudo make install

```
logan@logan-VirtualBox:~/kernel_build/linux-hwe-6.8-6.8.0$ make -j"$(nproc)"
  SYSHDR  arch/x86/include/generated/uapi/asm/unistd_32.h
  SYSHDR  arch/x86/include/generated/uapi/asm/unistd_64.h
  SYSHDR  arch/x86/include/generated/uapi/asm/unistd_x32.h
  SYSTBL  arch/x86/include/generated/asm/syscalls_32.h
  SYSHDR  arch/x86/include/generated/asm/unistd_32_ia32.h
  SYSHDR  arch/x86/include/generated/asm/unistd_64_x32.h
  SYSTBL  arch/x86/include/generated/asm/syscalls_64.h
  HYPERCALLS arch/x86/include/generated/asm/xen-hypercalls.h
  HOSTCC  arch/x86/tools/relocs_32.o
  HOSTCC  arch/x86/tools/relocs_64.o
  HOSTCC  arch/x86/tools/relocs_common.o
  HOSTLD  arch/x86/tools/relocs
  HOSTCC  scripts/genksyms/genksyms.o
  YACC    scripts/genksyms/parse.tab.[ch]
  HOSTCC  scripts/genksyms/parse.tab.o
  LEX     scripts/genksyms/lex.lex.c
  HOSTCC  scripts/genksyms/lex.lex.o
```

```
  INSTALL /lib/modules/6.8.12/kernel/net/hsr/hsr.ko
  SIGN    /lib/modules/6.8.12/kernel/net/hsr/hsr.ko
/bin/sh: 1: ./debian/scripts/sign-module: Permission denied
  INSTALL /lib/modules/6.8.12/kernel/net/qrtr/qrtr.ko
  SIGN    /lib/modules/6.8.12/kernel/net/qrtr/qrtr.ko
/bin/sh: 1: ./debian/scripts/sign-module: Permission denied
  INSTALL /lib/modules/6.8.12/kernel/net/qrtr/qrtr-smd.ko
  SIGN    /lib/modules/6.8.12/kernel/net/qrtr/qrtr-smd.ko
/bin/sh: 1: ./debian/scripts/sign-module: Permission denied
  INSTALL /lib/modules/6.8.12/kernel/net/qrtr/qrtr-tun.ko
  SIGN    /lib/modules/6.8.12/kernel/net/qrtr/qrtr-tun.ko
/bin/sh: 1: ./debian/scripts/sign-module: Permission denied
  INSTALL /lib/modules/6.8.12/kernel/net/qrtr/qrtr-mhi.ko
  SIGN    /lib/modules/6.8.12/kernel/net/qrtr/qrtr-mhi.ko
/bin/sh: 1: ./debian/scripts/sign-module: Permission denied
  INSTALL /lib/modules/6.8.12/kernel/virt/lib/irqbypass.ko
  SIGN    /lib/modules/6.8.12/kernel/virt/lib/irqbypass.ko
/bin/sh: 1: ./debian/scripts/sign-module: Permission denied
  INSTALL /lib/modules/6.8.12/kernel/ubuntu/ubuntu-host/ubuntu-host.ko
  SIGN    /lib/modules/6.8.12/kernel/ubuntu/ubuntu-host/ubuntu-host.ko
/bin/sh: 1: ./debian/scripts/sign-module: Permission denied
  DEPMOD  /lib/modules/6.8.12
logan@logan-VirtualBox:~/kernel_build/linux-hwe-6.8-6.8.0$ sudo make install
```

```
logan@logan-VirtualBox:~/kernel_build/linux-hwe-6.8-6.8.0$ sudo make install
  INSTALL /boot
run-parts: executing /etc/kernel/postinst.d/initramfs-tools 6.8.12 /boot/vmlinuz
-6.8.12
update-initramfs: Generating /boot/initrd.img-6.8.12
run-parts: executing /etc/kernel/postinst.d/unattended-upgrades 6.8.12 /boot/vml
inuz-6.8.12
run-parts: executing /etc/kernel/postinst.d/update-notifier 6.8.12 /boot/vmlinuz
-6.8.12
run-parts: executing /etc/kernel/postinst.d/vboxadd 6.8.12 /boot/vmlinuz-6.8.12
VirtualBox Guest Additions: Building the modules for kernel 6.8.12.

VirtualBox Guest Additions: Look at /var/log/vboxadd-setup.log to find out what
went wrong
run-parts: executing /etc/kernel/postinst.d/xx-update-initrd-links 6.8.12 /boot/
vmlinuz-6.8.12
I: /boot/initrd.img-6.8.12 is now a symlink to initrd.img-6.17.5
```

```
logan@logan-VirtualBox:~$ uname -r
6.8.12
```

4. Bonus

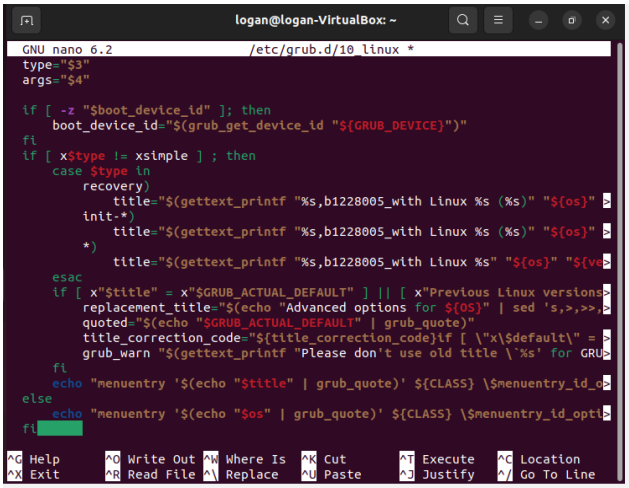
4.1 Modify the default kernel for booting (設6.17.5為預設開機)

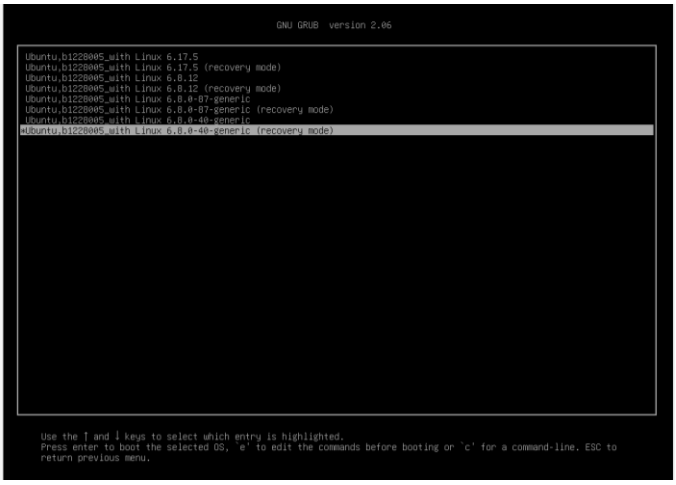
```
logan@logan-VirtualBox:~$ grep "menuentry '" /boot/grub/grub.cfg | nl
     1  menuentry 'Ubuntu' --class ubuntu --class gnu-linux --class gnu --class os $menuentry_id_option '
gnulinux-simple-a37a7e75-cb69-40ca-a0ed-f55665d7ba3b' {
     2          menuentry 'Ubuntu, with Linux 6.17.5' --class ubuntu --class gnu-linux --class gnu --clas
s os $menuentry_id_option 'gnulinux-6.17.5-advanced-a37a7e75-cb69-40ca-a0ed-f55665d7ba3b' {
     3          menuentry 'Ubuntu, with Linux 6.17.5 (recovery mode)' --class ubuntu --class gnu-linux --
class gnu --class os $menuentry_id_option 'gnulinux-6.17.5-recovery-a37a7e75-cb69-40ca-a0ed-f55665d7ba3b'
 {
     4          menuentry 'Ubuntu, with Linux 6.8.12' --class ubuntu --class gnu-linux --class gnu --clas
s os $menuentry_id_option 'gnulinux-6.8.12-advanced-a37a7e75-cb69-40ca-a0ed-f55665d7ba3b' {
     5          menuentry 'Ubuntu, with Linux 6.8.12 (recovery mode)' --class ubuntu --class gnu-linux --
class gnu --class os $menuentry_id_option 'gnulinux-6.8.12-recovery-a37a7e75-cb69-40ca-a0ed-f55665d7ba3b'
 {
     6          menuentry 'Ubuntu, with Linux 6.8.0-87-generic' --class ubuntu --class gnu-linux --class
gnu --class os $menuentry_id_option 'gnulinux-6.8.0-87-generic-advanced-a37a7e75-cb69-40ca-a0ed-f55665d7b
a3b' {
     7          menuentry 'Ubuntu, with Linux 6.8.0-87-generic (recovery mode)' --class ubuntu --class gn
u-linux --class gnu --class os $menuentry_id_option 'gnulinux-6.8.0-87-generic-recovery-a37a7e75-cb69-40c
a-a0ed-f55665d7ba3b' {
     8          menuentry 'Ubuntu, with Linux 6.8.0-40-generic' --class ubuntu --class gnu-linux --class
gnu --class os $menuentry_id_option 'gnulinux-6.8.0-40-generic-advanced-a37a7e75-cb69-40ca-a0ed-f55665d7b
a3b' {
     9          menuentry 'Ubuntu, with Linux 6.8.0-40-generic (recovery mode)' --class ubuntu --class gn
u-linux --class gnu --class os $menuentry_id_option 'gnulinux-6.8.0-40-generic-recovery-a37a7e75-cb69-40c
a-a0ed-f55665d7ba3b' {
    10  menuentry 'UEFI Firmware Settings' $menuentry_id_option 'uefi-firmware' {
logan@logan-VirtualBox:~$
```

```
logan@logan-VirtualBox:~$ uname -r
6.17.5
```

```
logan@logan-VirtualBox: ~
  GNU nano 6.2                    /etc/default/grub
# If you change this file, run 'update-grub' afterwards to update
# /boot/grub/grub.cfg.
# For full documentation of the options in this file, see:
#   info -f grub -n 'Simple configuration'

GRUB_DEFAULT="Advanced options for Ubuntu>Ubuntu, with Linux 6.17.5"
GRUB_TIMEOUT_STYLE=menu
GRUB_TIMEOUT=10
GRUB_DISTRIBUTOR=`lsb_release -i -s 2> /dev/null || echo Debian`
GRUB_CMDLINE_LINUX_DEFAULT="quiet splash"
GRUB_CMDLINE_LINUX=""
GRUB_TERMINAL_INPUT=at_keyboard

# Uncomment to enable BadRAM filtering, modify to suit your needs
# This works with Linux (no patch required) and with any kernel that obtains
# the memory map information from GRUB (GNU Mach, kernel of FreeBSD ...)
#GRUB_BADRAM="0x01234567,0xfefefefe,0x89abcdef,0xefefefef"

# Uncomment to disable graphical terminal (grub-pc only)
#GRUB_TERMINAL=console

# The resolution used on graphical terminal
# note that you can use only modes which your graphic card supports via VBE
# you can see them in real GRUB with the command `vbeinfo'
#GRUB_GFXMODE=640x480

# Uncomment if you don't want GRUB to pass "root=UUID=xxx" parameter to Linux
#GRUB_DISABLE_LINUX_UUID=true

# Uncomment to disable generation of recovery mode menu entries
#GRUB_DISABLE_RECOVERY="true"

# Uncomment to get a beep at grub start
#GRUB_INIT_TUNE="480 440 1"
                                        [ Read 34 lines ]
^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute     ^C Location    M-U Undo
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify     ^/ Go To Line  M-E Redo
```

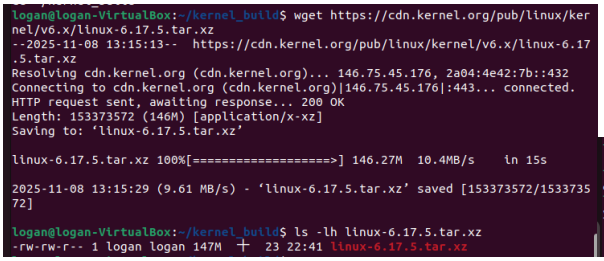
4.2 Modify the list of kernels in the GRUB menu(改名為%s,b1228005_with Linux %s)

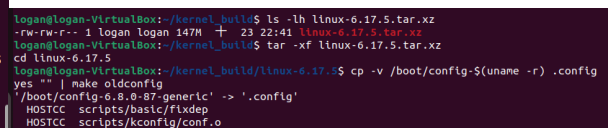




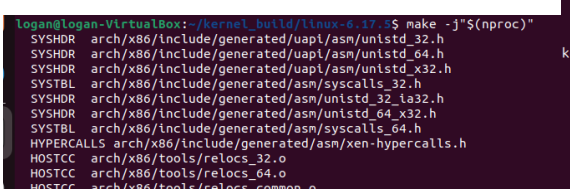
4.3.1Bonus:Download, compile and install Linux kernel 6.17.5

從官方 Linux kernel 主站（kernel.org） 下載 Linux 6.17.5 source code 並解壓縮





4.3.2編譯 Kernel,安裝模組,安裝kernel本體



```
logan@logan-VirtualBox:~/kernel_build/linux-6.17.5$ sudo make install
  INSTALL /boot
run-parts: executing /etc/kernel/postinst.d/initramfs-tools 6.17.5 /boot/vmlinuz
-6.17.5
update-initramfs: Generating /boot/initrd.img-6.17.5
W: Possible missing firmware /lib/firmware/i915/bmg_dmc.bin for built-in driver
i915
W: Possible missing firmware /lib/firmware/i915/xe2lpd_dmc.bin for built-in driv
er i915
W: Possible missing firmware /lib/firmware/i915/xe3lpd_dmc.bin for built-in driv
er i915
W: Possible missing firmware /lib/firmware/rtl_nic/rtl8127a-1.fw for built-in dr
iver r8169
W: Possible missing firmware /lib/firmware/rtl_nic/rtl8126a-3.fw for built-in dr
iver r8169
W: Possible missing firmware /lib/firmware/rtl_nic/rtl8126a-2.fw for built-in dr
iver r8169
W: Possible missing firmware /lib/firmware/rtl_nic/rtl8125bp-2.fw for built-in d
river r8169
W: Possible missing firmware /lib/firmware/rtl_nic/rtl8125d-2.fw for built-in dr
iver r8169
W: Possible missing firmware /lib/firmware/rtl_nic/rtl8125d-1.fw for built-in dr
```

```
logan@logan-VirtualBox:~$ uname -r
6.17.5
logan@logan-VirtualBox:~$
```

安裝的kernel

```
logan@logan-VirtualBox:~$ ls /lib/modules
6.17.5  6.8.0-40-generic  6.8.0-87-generic  6.8.12
logan@logan-VirtualBox:~$
```

5. 遇到的問題

1. 我一開始在編譯kernel之前有把我的oracle virtual box管理員裡的Ubuntu硬碟調到100GB，但是跑到一半還是跳出硬碟空間不足，上網查詢後是因為我還沒有調整Ubuntu裡面實際使用得空間。
2. 我在編譯kernel時編譯的比同學久很多，問了之後是因為他們把cpu數量開到4，結果我只有開到1。調整完後相比一開始真的快很多，從幾個小時變成1個小時就可以編譯完成。
3. 我找不到一開始對應的kernel的版本，所以我問ChatGPT，問它6.8.0的kernel的版本要去哪裡下載?以下為ChatGPT的回答。

Ubuntu 22.04 使用的是 “Hardware Enablement (HWE)” kernel 系列，
對應 6.8.0 的原始碼要用：apt source linux-hwe-6.8

6. 參考資料:

ChatGPT,
分配硬碟空間:http://www.j4.com.tw/comp-qna/ubuntu-%E6%96%B0%E5%A2%9E%E7%A1%AC%E7%A2%9F%EF%BC%88%E7%A3%81%E7%A2%9F%E5%88%86%E5%89%B2%E3%80%81%E6%A0%BC%E5%BC%8F%E5%8C%96%E8%88%87%E6%8E%9B%E8%BC%89%EF%BC%89/
ubuntu kernel安裝:
https://medium.com/@EricChou711/linux-kernel-driver-%E9%87%8D%E6%96%B0%E7%B7%A8%E8%AD%AF-for-ubuntu-fe6b016ef8fc